

TAMS Mock Election Results

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Abstract

The TAMS Mock Election is conducted by Ballot to encourage civic discourse at the Texas Academy of Mathematics and Science (TAMS) and to capture our school's unique political voice. The election emulated real-life conditions by adhering to the Texas Election Code's ballot order rules and using a First-Past-The-Post (FPTP) system. With 115 participants (about a third of TAMS students), the turnout was considered demographically representative of the student body. Key issues identified by voters included the economy, healthcare, and education, reflecting broader national concerns but with unique priorities specific to TAMS. The results showed that while Kamala Harris narrowly "won" the presidential vote, third-party candidates played a significant role, underscoring diverse political alignments within the TAMS community. Detailed analyses by gender and class revealed notable differences in issue prioritization, particularly around topics like abortion and national security. The mock election demonstrates the importance of political awareness even among those who aren't yet eligible to vote, highlighting potential improvements for future student-led electoral initiatives, such as experimenting with alternative voting systems like STV or STAR.

1 Prologue

Sometimes, we take for granted the importance of democracy and how individuals and groups have fought for the right to choose their government. However, making good use of that right is difficult. As John F. Kennedy said, "The ignorance of one voter in a democracy impairs the security of all". Now, being a math and science academy, civics might not be the most important thing in the world to a lot of you reading, and that's OK! But no matter if you read the news everyday, or just once a week in your inbox, it's important to stay up to date on current events so you can make your own decisions.

I say "make your own decisions" because without a proper understanding of the world around you, it's tough to actually cast your ballot as your voice. Especially in the 2 party system here, it's hard to find candidates sharing your beliefs and to have faith in them.

Sometimes it leads to this question: "If I'm only voting for the lesser of two evils, why should I vote? Democracy is failing," and the answer is, that's what all the dictators in the world want you to think. Millions have died for democracy and freedom, and when that right is gone, it's gone.

And the other thing which is asked: "Is this election fair?", and the answer is no, it is not fair. If you want evidence for that, they are listed in the references. [Bogdanor \[1997\]](#) [Edwards \[2023\]](#) [Kennedy \[2008\]](#) However, it's the best thing we've got right now, so you might as well use it. As bad as it might be, it at least gives you the chance to change your leaders.

So, in the interest of democracy, Ballot has the TAMS Mock Election. Obviously, most of us can't vote but there's definitely ways that we can be civically engaged, and this is our way of doing so. (Or just me projecting my own desire to vote as a psychological defense mechanism to everyone at TAMS, take it how you wish.) Our last turnout at the midterms was 59, but this year's turnout was almost double at 115! (And that's an exclamation, not a factorial sign). I have to give a truly heartfelt thank you to everyone who voted, you made this possible.

- Archith Sharma, Election Administrator

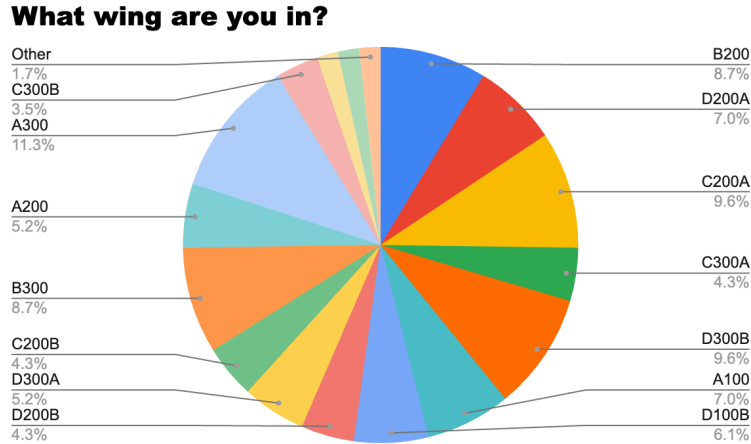


Figure 1: Voter Distribution

2 Methodology

There were two things I could have done, make the election an accurate simulation or make them fairer. Unfortunately, those things are mutually exclusive. The Texas Election Code dictates that candidates “shall be arranged in descending order of the number of votes received statewide by each party’s candidate for governor in the most recent gubernatorial general election, beginning on the left with the party whose candidate received the highest number of votes.” In non-legalese, that means “Whoever gets the most votes for governor is listed first on the ballot, and then down from there.” Research has shown that name-order effects showed up in almost half of all races and gave a candidate a 2.5% advantage, which is A LOT! [Legislature \[1985\]](#) [Miller and Krosnick \[1998\]](#) Obviously, to make it fair, I could have randomized it, but I chose to make it a simulation of reality.

The next thing I could have done to make it fair is obviously use STAR (Score then Automatic Runoff) or STV (Single Transferable Vote) voting. Once again, for the sake of simplicity on the ballot and to accurately simulate the election, I (begrudgingly) made all of the ballots First Past the Post, or just most votes wins. Honestly, STV would yield similar results as FPTP, though STAR would have intrigued me based on approval rates. Next election, this is one of the things that we could look to try.

Don’t forget about STV (aka ranked-choice voting, instant runoff) though! I used it in our fun question, and the results show a limitation of that system. I know we all love science, so stay reading until the end for those results.

If you want to play around with our data, it is [uploaded to my GitHub page, along with some code I used](#).

3 Voters

Now, before the moment you’ve been waiting for, I’ve added insights into our voter base. I’ve also attached some commentary here, to show how things have changed from last time and how TAMS follows or bucks national patterns.

3.1 Voter Summary

Voters were more or less uniform across wings and floors (Figure 1), so I’m statistically confident in saying this a representative sample for all of TAMS.

3.2 Voting Issues

People were asked to pick 3 important voting issues in the election. The top 5 voting issues were:

What three voting issues are most important to you?

 Copy chart

115 responses

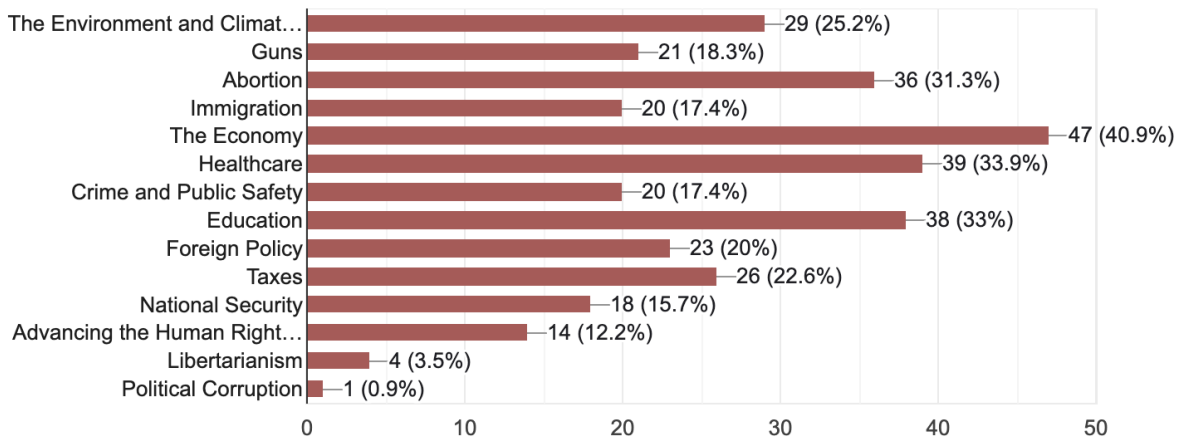


Figure 2: Voting Issues

1. The Economy
2. Healthcare
3. Education
4. Abortion
5. The Environment and Climate Change

in that order. The 2 most important observations I had from this and the midterms were:

- **Abortion stays important.** It's a strong issue of Harris's campaign, and it's something she has pushed as the reason to vote for her. I'm not surprised that abortion is this high on the list. Also, it's more important for women than men. 60% of the people saying abortion were one of the top three issues were women.
- **The environment takes a backseat to the economy.** One of the more surprising finds that were in the data were the amount of people talking about the economy as an issue. Ranked #7 in the midterms, it made a massive jump to being TAMS' most important issue. In the last two years, home prices have shot up to record highs, and energy prices have been moving up as well.

This influx in importance to the economy also led to one issue being pushed down: the environment. While traditionally, Dems say they are for environmentalism while Republicans has been pro-oil business (though really, they are all pro-oil business and not environmentalists) the environment hasn't been made a significant issue. As Harris has tried to court Republican moderates, she's cut back on climate rhetoric, making it less important in the election as well - in a nutshell, it's tough to rationalize the planet as an important issue if you can't pay for anything on it.

- **Education and Healthcare were both continuities from 2022**, and TAMSters continue to value those two - most likely because of two reasons. A lot of us want to go into healthcare, so we rate it highly. And, everyone who came to TAMS came on the merit that we value education and want to learn more at UNT. So, I'm guessing that's why we rate education highly.

3.3 Parties

Here are some of the findings for people by party:

- Republicans' most important issues are guns and the economy.

- Democrats' most important issues are healthcare and abortion.
- Everyone generally believes the Democrats are the best party. Libertarians and Greens at TAMS both lean Democratic if they had to choose via the STV plot (Figure 3 and 4).

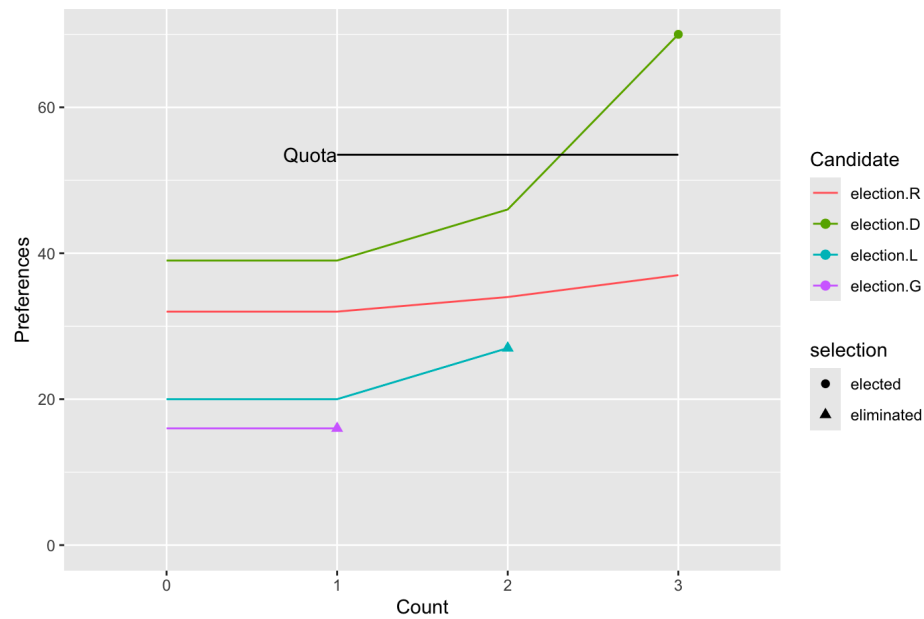


Figure 3: Party STV Plot

Results of Single transferable vote

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=====
Number of valid votes: 115
Number of invalid votes: 0
Number of candidates: 4
Number of seats: 1
```

	1	2-trans	2	3-trans	3
Quota	57.501		57.501		57.501
election.R	35.000	2	37.000	4	41.000
election.D	43.000	7	50.000	24	74.000
election.L	21.000	7	28.000	-28	
election.G	16.000	-16			
Elected				election.D	
Eliminated	election.G	election.L			

Elected: election.D

Figure 4: Transfer of votes

4 Election Results (sorry about the colors being confusing)

In this section, the analysis by age and gender will only be containing the votes for president and U.S. senator due to lack of sample size in those who filled out the rest of the ballot. You can get the gender and class analysis for every race by [downloading the code in my GitHub though, just follow the comments.](#)

4.0.1 The General Election

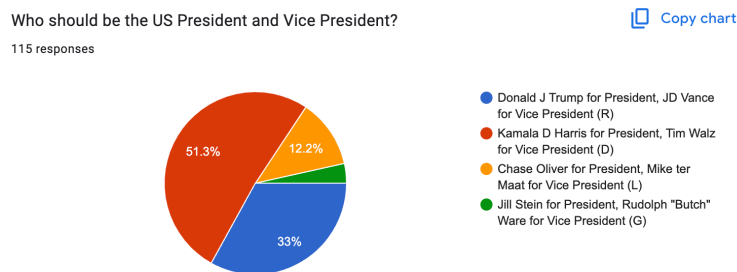


Figure 5: Presidential Election

Kamala BARELY scraped a majority! Honestly, very surprised. I expected votes for her to be in the 55-65% Harris range initially, but the turnout of third-party and Trump voters killed her majority.

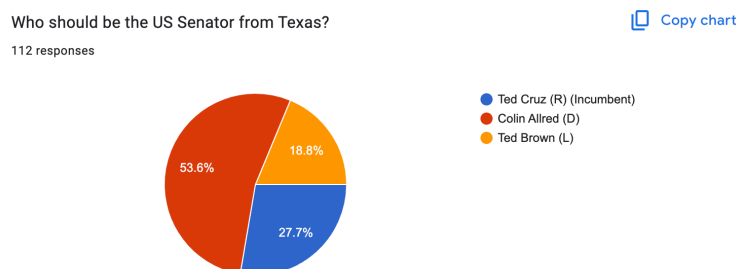


Figure 6: Senate Election

For the senate, lots of third-party voters once again. Libertarian voters generally preferred democrats over republicans, however so in a hypothetical runoff, Allred would still win.

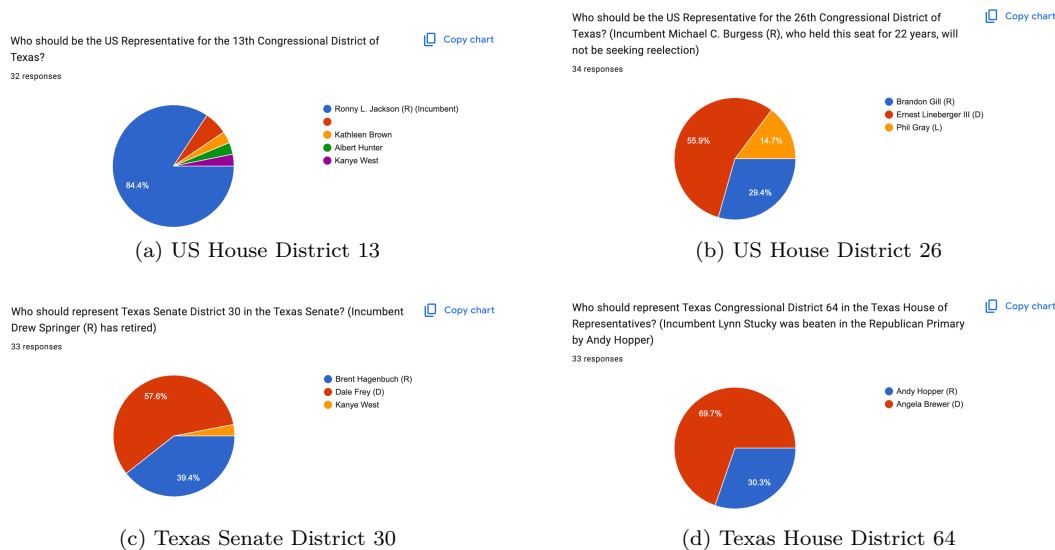


Figure 7: Multiple Congressional Districts

And as for the congressional races, not much insight from here as not many people filled the full ballot.

4.1 Analysis by Gender & Class

This analysis will be done just for the US Senate and Presidential Race. Once again, you can replicate it by [following my GitHub](#). I know that picking the gender and class based on wing is susceptible to error, but I'd say they cancel each other out on aggregate (juniors in senior wings, senior mentors, etc.). There were 50 seniors and 65 juniors, 62 guys and 53 girls. There were 33 junior girls, 32 junior guys, 21 senior girls, and 29 senior guys. The sample is mostly representative although senior girls are slightly underrepresented. Here is the party preference by gender and class, along with the election results:

	D	R	L	G
Seniors	1.86	2.66	2.72	2.76
Juniors	2.17	2.51	2.71	2.62
Guys	2.10	2.53	2.61	2.76
Girls	1.96	2.62	2.70	2.72

Table 1: Average party ranking by class or gender

Choice	Girls	Guys
Chase Oliver (L)	3	11
Donald J Trump (R)	19	19
Jill Stein (G)	1	3
Kamala D Harris (D)	30	29

Table 2: Presidential Choices by Gender

Issues	Guys Frequency	Girls Frequency
Abortion	16.1	49.1
Advancing the Human Rights of LGBTQI+ Persons	12.9	11.3
Crime and Public Safety	16.1	18.9
Education	30.6	35.8
Foreign Policy	25.8	13.2
Guns	21.0	15.1
Healthcare	27.4	41.5
Immigration	17.7	17.0
Libertarianism	1.6	5.7
National Security	21.0	9.4
Political Corruption	1.6	0.0
Taxes	30.6	13.2
The Economy	48.4	32.1
The Environment and Climate Change	24.2	26.4

Table 3: Frequency of Issues by Gender

Choice	Girls	Guys
Colin Allred (D)	31	29
Ted Brown (L)	7	14
Ted Cruz (R) (Incumbent)	13	18
No vote	2	1

Table 4: Senate Choices by Gender

And my commentaries:

- **Abortion is three times as important to girls to guys, while taxes, national security and the economy are 50-100% more important to guys.** Honestly, expected. I know that abortion affects guys as well - women not having the right to choose burdens healthcare systems. However, you can see why the issues the guys picked are important as well. The economy and taxes are really important, especially in this era of uncertainty.
- **Gender wasn't indicative of political preference.** I expected girls to be far more blue then they actually were. If I were to sample out of the guys and girls, I couldn't tell who based on their ballot.

Here is the analysis by class. Here are the important issues, and their voting choices.

Choice	Seniors	Juniors
Chase Oliver (L)	3	11
Donald J Trump (R)	12	26
Jill Stein (G)	4	0
Kamala D Harris (D)	31	28

Table 5: Presidential Choices by Class

Issues	Senior Frequency	Junior Frequency
Abortion	42	23.1
Advancing the Human Rights of LGBTQI+ Persons	16	9.2
Crime and Public Safety	16	18.5
Education	40	27.7
Foreign Policy	24	16.9
Guns	16	20.0
Healthcare	34	33.8
Immigration	20	15.4
Libertarianism	2	4.6
National Security	10	20.0
Political Corruption	0	1.5
Taxes	16	27.7
The Economy	32	47.7
The Environment and Climate Change	28	23.1

Table 6: Frequency of Issues by Class

Choice	Seniors	Juniors
Colin Allred (D)	33	27
Ted Brown (L)	8	13
Ted Cruz (R) (Incumbent)	9	22
No vote	3	0

Table 7: Senate Choices by Gender

And my commentaries:

- **Seniors were far more blue than juniors.** Likely due to the makeup of seniors that voted. I doubt that the senior polling was too representative since the larger junior sample was closer to the center politically. In addition to that, it led to similar patterns in the voting issues that were found in the gender column, although juniors still did value healthcare.
- **Libertarians.** The Junior class had far more Libertarian voters, though still generally proportional. Needs a bit more research as I can't really pinpoint why as of this moment.

5 TAMS Election (RCV and the insane way I could win)

Congrats, Srinjoy (second place vote merchant). Now comes the question, do I let my self-interest outweigh the greater good by switching to FPTP?

```
Results of Single transferable vote
=====
Number of valid votes:  90
Number of invalid votes: 25
Number of candidates:   5
Number of seats:        1

|          | 1| 2-trans| 2| 3-trans| 3| 4-trans| 4|
|:-----|:-----|:-----|:-----|:-----|:-----|:-----|
|Quota    | 45.001|      | 44.501|      | 44.501|      | 44.501|
|Adithi   | 17.000| 1| 18.000| 5| 23.000| -23|      |
|Srinjoy   | 20.000| 4| 24.000| 10| 34.000| 13| 47.000|
|Tracy     | 17.000| 1| 18.000| -18|      |      |      |
|Hunter    | 11.000| -11|      |      |      |      |      |
|Archith   | 25.000| 4| 29.000| 3| 32.000| 10| 42.000|
|Tie-breaks|      |      | fo|      |      |      |      |
|Elected  |      |      |      |      |      |      | Srinjoy|
|Eliminated| Hunter|      | Tracy|      | Adithi|      |      |

Elected: Srinjoy
```

Figure 8: RCV for all of Ballot

The interesting part here though, is how I could win. If Tracy and Adithi had 6 more votes, they'd eliminate Srinjoy in the second round. If I was voting and trying to get myself elected, the mathematical best way to do that is put myself third! [Deshpande et al. \[2024\]](#) This spoiler effect is definitely worse in FPTP than RCV, but less exploitable in STAR. See this video for further details. [Muller \[2024\]](#)

6 Thank you!!

Thank you for voting and reading until the end. Your interest in the mock election means a lot to me. To explore the data and code I used, check out [my Github page](#) for this, as promised.

Hope you enjoyed,
Archith Sharma, Election Administrator

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